

RNA co-Immunoprecipitation

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Day 0: Plate Cells (for cos-1 cells, split cell ~8hr before transfection)

- COS-1 cells: 6e5 cells/poly lysine-coated 60m plate; 293T: 2e6 cells/60mm plate.
- Use 4% lysate for TCL IP ($=4.72 \times 10^4$ cell equivalent, 25 μ L out of 620.5 μ L lysate)

Day 0: Transfect Cells

8 hours after plating cells, transfect each plate with desired plasmid(s).

Before starting:

- Label 1 1.5 mL Eppies per sample.
 - Thaw DNA at 55°C in Thermomixer and rotate at >1000 RPM for 15 minutes to resuspend.
 - Thaw PEI (MANUFACTURER, CATALOGUE NUMBER) at 55°C in Thermomixer and rotate at >1000 RPM for 15 minutes to resuspend.
1. For each transfection condition, dilute DNA in 500 μ L pre-warmed, serum-free medium.
 2. Let sit 5 minutes at room temperature (RT).
 3. Add 20 μ L of pre-warmed PEI (more than 15min at 55°C) to DNA/ serum-media, incubate at RT 15 minutes.
 4. Change media on cells to 3 mL of transfection media.
 5. Add transfection mix dropwise to cells while gently rocking the plate.

Day 1: Change media (when you culture cells more than 24 hrs after transfection)

Before starting:

- Warm PBS + MgCl₂/CaCl₂
- Warm culture media (for 293Ts and Cos-1, 500 mLs DMEM + 10% FBS + 5 mLs penicillin/streptomycin)

~14 hours post transfection, rinse cells twice with 3 mL warmed PBS containing CaCl₂ and MgCl₂ to remove untransfected DNA, then add 3.5 mL fresh media.

Day 2: HARVEST VLPs and lysate

For COS-1 cells, lysate and media should be harvested 34-38 hours post-transfection.
For 293T cells, lysate and media should be harvested 15-18 hours post-transfection.

Harvest VLPs (isolating VLPs from other particles in the supernatant):

Before starting:

- Label and pre-chill 2 1.5 mL Eppies and 1 15 mL Falcon tube per sample.
- Pre-chill benchtop Allegra centrifuge to 4°C.
- Prepare 1.5 mLs of 30% sucrose per sample.
- Make up 500 μ L 1x NN+M with 1:100 Protease inhibitor (PI, MANUFACTURER, CATALOGUE NUMBER) and 1:1000 RNaseOUT per sample (MANUFACTURER, CATALOGUE NUMBER), chill (see end of protocol).
- Obtain a 1 mL syringe and a 20-gauge needle for loading the sucrose cushion.

1. Remove a plate of cells from the incubator and place on ice.
2. Collect supernatant in to 15mL conical tube * (harvest corresponding cells in NN+M before moving on to next step, see below).
3. Spin supernatant at 2000 RPM for 7 min at 4°C.
4. Filter supernatant through 0.45µm pore size filter (0.22 µM will filter out your virus!!)
5. Add 2.7mL of the filtered supernatant into a WHAT SIZE, (MANUFACTURER, CATALOGUE NUMBER) ultracentrifuge tube.
6. Draw up 1.5 mL of 30% (w/v) sucrose into your 1 mL syringe.
7. Place the tip of the needle at the bottom of the ultracentrifuge tube and slowly layer the sucrose underneath the filtered supernatant.
8. Centrifuge at 60K RPM for 30 min at 4°C.
9. Suction the media and sucrose, tilting the tube slowly as you aspirate and being careful not to disturb the pellet at the bottom of the tube (which may not be visible).
10. Add 100 µL 30% sucrose.
11. Vortex tube to resuspend the pellet and collect this VLP fraction into 1.5mL Eppendorf tube.

Harvest lysate in NMPH (dissociating ribosomes and releasing the nascent chain):

Before starting:

- Label and pre-chill 3 1.5 mL Eppie tubes per sample.
 - Pre-chill benchtop Allegra centrifuge to 4°C.
 - Pre-chill a microcentrifuge to 4°C.
 - Pre-chill 1 mL syringes with 20 gauge needles attached.
 - Prepare a “cell scraper” – an aspirator pipette with a rubber policeman scraper at the end.
 - Thaw an aliquot of Puromycin (need 26.5 µL for each sample) – Puromycin should not undergo multiple freeze-thaw cycles.
 - Make sure you have 144 µL of 2M KCl NN+M (high salt) per sample, keep chilled.
1. Rinse cells with 3mL ice-cold PBS containing CaCl₂ and MgCl₂ twice to remove untransfected DNA.
 2. Add 500 µL of 1x Lysis Buffer NN+M (+ 1:100 Protease Inhibitor and 1:1000 RNaseOUT) per plate, scrape lysate, transfer to chilled tube.
 3. Shear lysed cells 20-times through a chilled 20-guage needle.
 4. Spin lysed cells at 1000 RPM for 10 minutes at 4°C in tabletop Allegra, then move supernatant to a new tube.
 5. Spin supernatant at 14k RPM 1 minute 4°C in microcentrifuge.
 6. Transfer 450 µL of lysate to a new tube.
 7. Warm up lysate and Puromycin for 5 minutes at 26°C.
 8. Add 26.5 µL Puromycin to 450 µL of lysate at final concentration of 1mM (stock concentration of Puromycin is 18mM)
 9. Incubate for 10 minutes at 26°C.
 10. Add 144 µL 2M high salt NN+M at final concentration 0.5M KCl (620.5 µL lysate final volume).
 11. Incubate for 10 min at 26°C
 12. Place tubes on ice for at least 10 minutes.
 13. Centrifuge at max speed for 1 min in a 4°C microcentrifuge.

Proteinase K reaction and TRIZOL (stripping proteins and preparing RNA for isolation):

Before starting:

- Make up 1.5 mL Eppie tubes for TCL and VLP from each sample.
- Make up non-detergent 1x NN buffer (see end of protocol).

- Obtain 1 mL Trizol (Ambion, cat# 15590026) for each sample (make an aliquot of the needed volume + 10%, being careful not to draw from the bottom of the bottle).
- Preheat RNA bench heat-block or water bath to 55°C.
- Obtain 20 mg/mL Proteinase K (Sigma P4850).

1. Make up Proteinase K elution buffer

To make 1 ml (for approx. 10 samples):

- 975 µL of non-detergent 1x NN (see end of protocol)
- 15 µL Proteinase K
- 10 µL 10% SDS

2. a) TCL input: take 2.5 µL TCL out of total of 620.5 µL.
b) VLPs: Take 2.5 µL VLPs out of total ~100 µL and mix with 7.5 µL 1xNN buffer.
3. Add Proteinase K elution buffer to 100 µL final volume.
4. Incubate at 55°C for 30 minutes.
5. If water bath was used for incubation, dry tubes thoroughly.
6. Add 1 ml TRIZOL. Invert to mix well and incubate at RT for 5 minutes.
 - Store at -80C if stopping.

If you want to stop the experiment, stop at this point. RNA is not stable in isopropanol.
Next stop point is the end of cDNA reaction.

Day 3: RNA isolation and Reverse Transcription reaction

RNA isolation with TRIZOL:

Before starting RNA isolation and DNase treatment:

- Obtain 5 µL Glycoblue (Invitrogen AM9515 or AM9516) per sample.
- Obtain 100 µL BCP (BP 151 phase separation reagent, Molecular Research Center, CATALOGUE NUMBER) per sample.
- Prepare 0.7 mL of ice cold isopropanol per sample (store at -20°C until ready for use).
- Prepare 1 mL of ice cold 75% ethanol per sample with PCR grade or DEPC-treated water (store at -20°C until ready for use).
- Prepare 51 µL of 1xDNase buffer (Invitrogen cat# 18068-015) prepare with DEPC water) containing 1:1000 RNaseOut and 2U (1 µL) DNase (Invitrogen cat# 18068-015) per sample.
- Heat a clean-room heat block or water bath to 25°C.
- Heat a clean-room heat block or water bath to 65°C.
- Obtain 1.5 µL of 25mM EDTA per sample.
- Pre-chill an RNA-designated centrifuge.
- Clean benchtop, Pipetmen, and all equipment with RNase-ZAP (MANUFACTURER, CATALOGUE NUMBER) or similar reagent.

1. If samples are frozen, thaw at room temperature and set on ice. Invert many times to promote lysis of material.
2. Spin down briefly to collect inverted liquid.
3. Add 75 µg of glycoblue (5 µL) and mix well by inverting tube.
4. Spin down briefly to collect inverted liquid.
5. Add 100 µL of BCP.
6. Vortex for 10-15 seconds until emulsion forms, repeat 2 times.
7. Let sit on ice for 2-3 minutes.

8. Spin 12,000xg 15 minutes at 4°C in chilled RNA-designated microcentrifuge.
9. While tubes are spinning, for each sample transfer 0.7 mL of ice cold isopropanol to a fresh labeled tube. Set tubes on ice.
10. With a 1 mL Pipetman, transfer ~0.6 mL of the aqueous phase from the spun sample directly to the tube with the isopropanol. Keep all samples on ice.
11. Mix well by inverting at least 5 times.
12. Transfer to -20°C for at least 1 hour.
13. Spin at full-speed (14K RPM) for 30 minutes in cold room or in chilled benchtop centrifuge, then carefully use a 1 mL pipette to remove the supernatant without touching the pellet (tip the tube).
14. Wash pellet with 1 mL of 75% ice cold EtOH by inverting several times.
15. Spin full speed (14K RPM) for 10 minutes in chilled benchtop centrifuge.
16. Aspirate most of the EtOH.
17. Spin full speed (14K RPM) for 1 minute in chilled benchtop centrifuge, then pipet to remove the remaining EtOH (with P200).

18. MOVE samples to clean room: everything will be in a clean room or clean area after this

19. Allow RNA pellets to dry with open lids for a few minutes (~3). As pellets dry, close tube to prevent over-drying.

DNase I treatment (@ clean room):

1. Add **51 µL of 1xDNase buffer** (made with DEPC water) **containing 1:1000 RNaseOut and 2U DNase (1 µL)**, by pipetting bead of water on sidewall and letting it slide down to pellet. Be sure pellet ends up in water, sometimes they can end up on sidewall (static effect).
2. Tap samples to resuspend pellet. Spin down briefly.
3. Finish mixing with pipet 5 times, running the bead of water over tube sidewall 5 times while doing this to ensure the entire RNA pellet resuspends.
4. Incubate tubes at 25°C for 15 minutes.
5. Add 1.5 µL of 25mM EDTA, mix well, spin.
6. Heat sample at 65°C for 10 mins to inactivate DNase I (follow the manufacture protocol).

RT reaction to make cDNA from RNA (in a clean room or clean area):

Before starting:

- Pre-heat a water bath to 42°C.
- Pre-heat a heat block to 85°C.
- Label 1.5 mL Eppies for one +RT and one –RT control reaction per sample, plus a single no-RNA control.

1. Make cDNA master mix (iScript Advanced cDNA synthesis kit, 172-5037, BIO-RAD), including one no-RNA control (total).

+RT samples and – RNA control rxn (one total)

	x1
PCR grade H2O	10 µL
5X Script reaction mix	4 µL
20X RT enzyme mix	1 µL
(Sample: RNA	5 µL)
total:	15 (20) µL

2. For each sample, aliquot 15 μL master mix into Eppie tubes.
3. Add 5 μL of RNA to each Eppie tube (total 20 μL per reaction for each duplicate).
4. For RNA negative control: aliquot 15 μL master mix into Eppie tube.
5. Add 5 μL DEPC H₂O to this tube: (for no-RNA control to determine the purity of the cDNA reagents)
6. Make cDNA master mix for RT- controls (iScript Advanced cDNA synthesis kit, 172-5037, BIO-RAD), including one control per sample.

RT- controls to test the purity of the RNA

	x1
PCR grad H ₂ O	15 μL
(Sample: RNA	5 μL)
total:	15 (20) μL

7. Aliquot 15 μL into Eppie tubes.
8. Add 5 μL RNA to each tube (these are the RT- controls, which measure the DNA contamination in the RNA samples).

RT reaction

9. Incubate samples at 42°C for 30 minutes in the preheated water bath.
10. Incubate samples at 85°C for 5 minutes in the preheated heat block.
11. **Dilute cDNA 1:3** (transfer to new tubes – add 51 μL DEPC H₂O + 17 μL cDNA) – these can be stored at -80°C; but better to immediately proceed to qPCR if you can.

qPCR oligos:

- Oligo information is available in methods.
- When oligos arrive, dilute to 200 μM .
- Mix 200 μM forward and reverse oligos at 1:1 to make 100 μM stock.
- Before use, make 10 μM oligo mix.

qPCR standard for VIB or 1522 genome:

NOTE: always work with standards OUTSIDE of the clean room or clean area!!!

- Initial standard dilution is 1e8 copies/2 μL – make many 20 μL of aliquots of this dilution, stored at -80C.
- Thaw each aliquot only once, use freshly thawed aliquot for each dilution series
- Prepare standards at 1:10 dilutions ranging from 1e1-1e8 copies/2 μL (2 μL =per well), duplicated – add 5 μL each dilution to 45 μL DEPC water to make the next dilution
- Dilution series are only one time use, do not re-freeze for another use.

qPCR plate setup:

All the qPCR reagents are stored in the clean room, and this is where aliquoting is done. Standards are set up outside the clean room.

Before starting:

- Make up 2 25 μL qPCR reactions per sample – set up in clean room.
- Prepare your standard curve (see above) – set up outside the clean room.

- Prepare DEPC water (in the clean room boxes, aliquots made in the hood).
- Thaw Sybr green (MANUFACTURER, cat# 170-8884) and keep on ice, protected from the light.
- Obtain or make up 10 μ M oligos.
- NOTE: duplicate qPCR reactions are set up in Eppies first, then aliquoted to plate (include 16 wells of master mix below for duplicate standard curve to after sample qPCRs have been set up).
- Obtain a 96-well plate.

qPCR mix:

	x1
DEPC	9.75 μ L
2xSYBR	12.5 μ L
10 μ M oligos mix	0.75 μ L
(Sample: 4 fold diluted cDNA	2 μ L)
total:	23 (25) μL

1. Make up duplicate reactions, then aliquot into a clean 96-well plate, being careful not to introduce bubbles. Wells for standards should be in the top two rows, stacked. These will be completed outside the clean room.
2. Add 2 μ Ls of diluted cDNA to each non-standard sample well.
3. Place paper towel over plate in clean room, move to another bench to add prepared standards.
4. Load 2 μ L of each standard per well in duplicate.
5. Seal cover carefully with a foil from the RNA bench drawer, using a sealer from the same drawer to ensure a tight seal. Pay careful attention to the edges.
6. Centrifuge plate briefly to 500 RPM in a tabletop Allegra.
7. Plate is now ready to be set in qPCR machine, with well A1 on the top left and the notch at the bottom left.

qPCR program setting (for gRNA analysis)

1. 95°C, 30 seconds
2. 95°C, 15 seconds
3. 60°C, 15 seconds
4. 72°C, 30 seconds +10 sec for data collection
5. Back to 2, 35 cycles
6. 55°C, 1 min
7. 55°C- T_m measuring

BUFFER RECIPES (below):

Lysis buffers:

Tris is helpful if you need to add calcium or manganese, b/c these will precipitate phosphate (PBS); otherwise, phosphate is preferable. But need to have a good buffering system (Tris pH 7.4) for colPs.

10X NP40 Normal salts + MgCl₂ (NN+M)

0.1M	Tris Hcl (pH 7.4)
1.0M	NaCl
0.5M	KCl
10mM	MgCl ₂
6.25%	NP40

1X NP40 Normal salt + MgCl₂ (NN+M high salt 2M KCl)

10mM Tris HCl (pH 7.4)
0.1M NaCl
2M KCl
1mM MgCl₂
0.625% NP40

IP Wash buffers: USE 1xNN buffer set for NMPH harvested lysate

***Use 1x NN non-det. buffer for WB IP wash**

***Use 1x NN non-det. IP wash (Prot.K –compatible) for RNA analysis wash**

1x NN IP wash buffer	1x NN Non-det. IP wash buffer	1x NN non-det. IP wash (Prot.K –compatible)
10mM Tris-HCl, pH 7.4	10mM Tris-HCl, pH 7.4	10mM Tris-Ac, pH 8.0
0.1M NaCl	0.1M NaCl	0.1M NaCl
50mM KAc	50mM KAc	
0.625% NP40	100mM Tris Ac , pH 8.0	